

Advanced VLM Tasks

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Inspiring Excellence



Overview

- Core NLP Tasks
- Transformer Models
- LLMs
- BERT
- GPT & Llamas
- how to use LLMs via APIs (Groq)
- Metrics: BLEU, ROUGE, perplexity

Core NLP Tasks

1. Classification
2. Summarization
3. Question Answer
4. Text Generation
5. Machine Translation
6. Named Entity Recognition

Core NLP Tasks

NER Named Entity Recognition:

Identifying and classifying entities (like names, places, organizations, dates) in text

Sentence: “Elon Musk founded SpaceX in California in 2002.”

NER output:

Elon Musk → Person

SpaceX → Organization

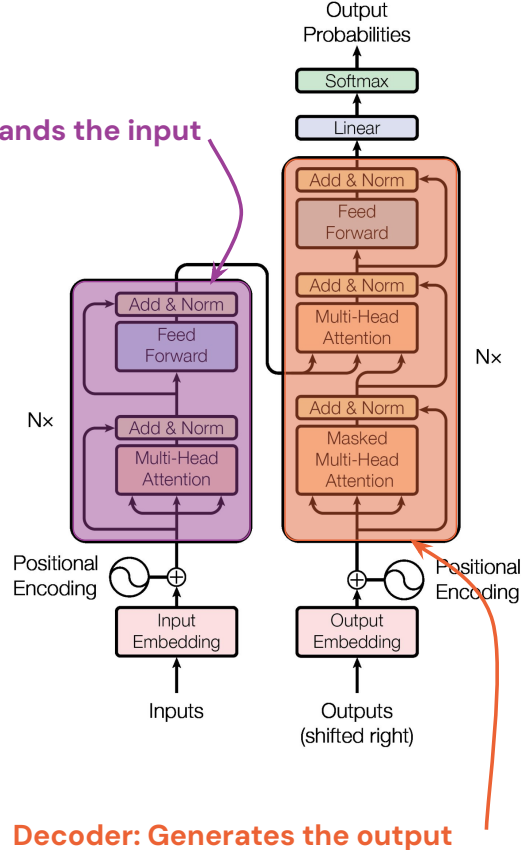
California → Location

2002 → Date

Transformers: Types?

Encoder-Only	Decoder Only	Encoder-Decoder
Focuses on understanding text	Focuses on generating text	Handles both understanding & generation
Examples: BERT, RoBERTa	Examples: GPT, LLaMA	Examples: T5, BART
Use cases: NER, sentiment analysis, other classification tasks	Use cases: chatbots, story writing, code generation	Use cases: translation, summarization, Q&A

Encoder: Understands the input



Transformers: How do we use it?

Usually, Transformer based models especially LLMs are very hard to train due to resources constraints. So, we:

Pretrained Models: Use models like BERT, GPT, RoBERTa, T5 for NLP tasks without training from scratch

Fine-tuning

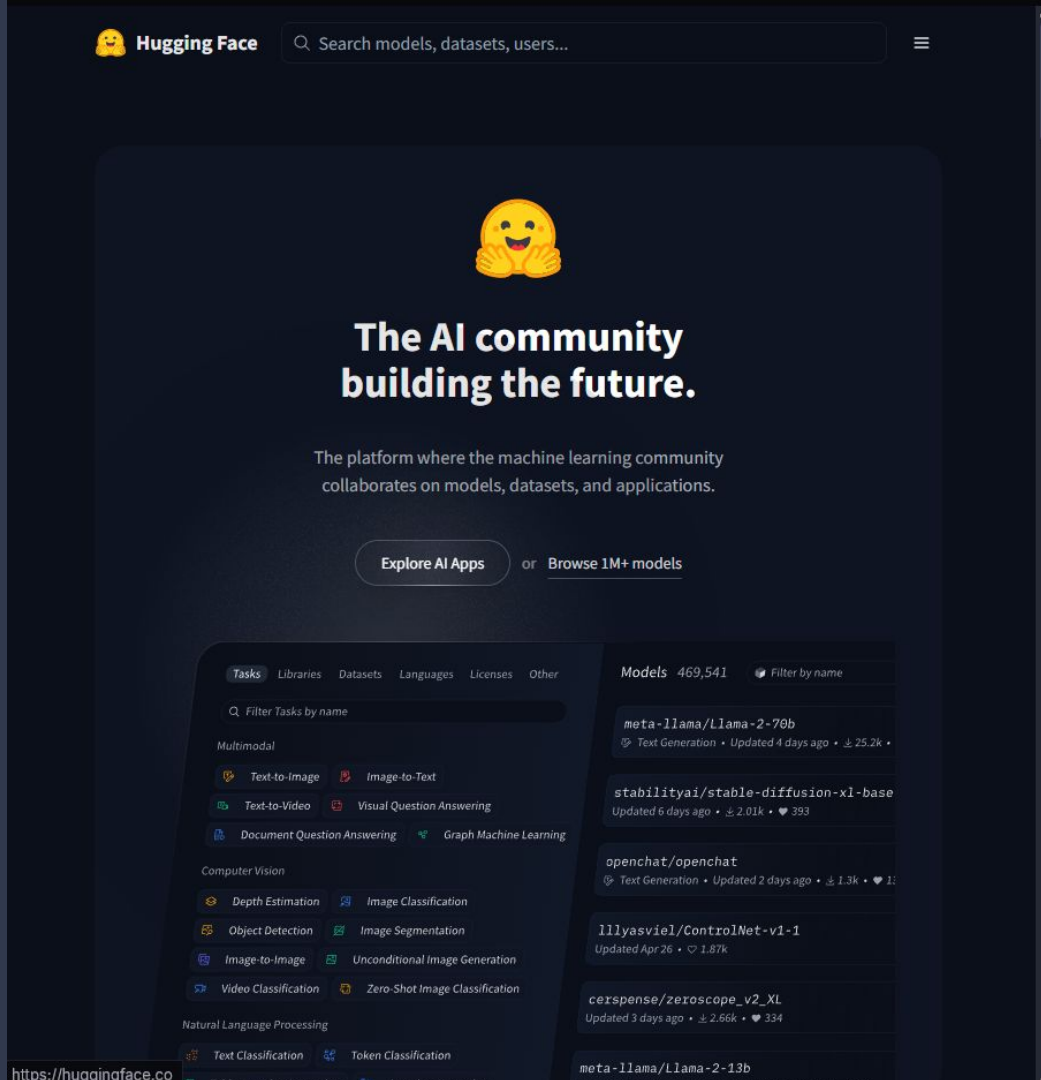
Adapt pre trained models for specific tasks (e.g., sentiment analysis, NER, summarization).

APIs & Libraries

Hugging Face Transformers, OpenAI, Groq, Cohere APIs

Hugging Face Transformers

- Provides thousands of pretrained models for NLP, vision, and speech
- Easy to use with just a few lines of code for training, fine-tuning, or inference
- Supports popular models (BERT, GPT, T5, LLaMA) with an active open-source community.



Hands on

How to use models from hugging face

How to fine tune models from hugging face

LLMs: GPT, LLama and others

GPT (OpenAI): Known for ChatGPT

LLaMA (Meta): Open-weight family, efficient and strong on many benchmarks, widely used for fine-tuning

Anthropic's Claude: Focus on safety, long-context reasoning, structured outputs

Mistral: Open-weight models optimized for speed and efficiency (example: Mixtral MoE)

And many more...

Current Trend: Ecosystem moving toward *open + efficient* models

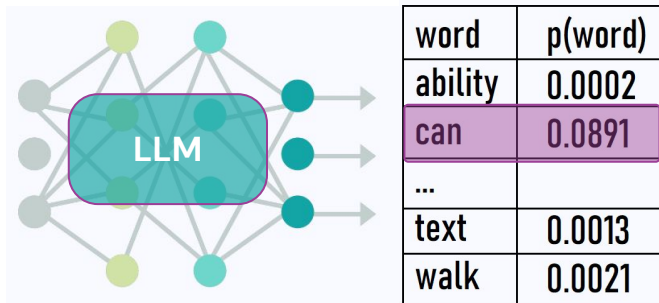
HOW DOES GPT WORK?



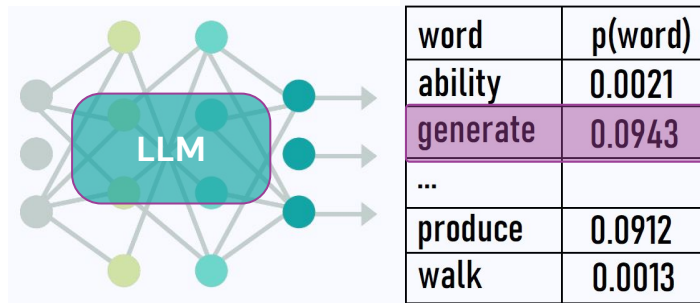
After training:

We can generate text by predicting one word at a time (Autoregressive)

[A trained language model]



[A trained language model **can**]



[A trained language model can **generate**]

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Using LLMs (APIs & Providers)

Direct APIs: Most LLMs are accessed via REST endpoints or Python SDKs.

Popular Providers:

OpenAI: GPT-4/4o for chat, embeddings, fine-tuning

Google Gemini API: For google Gemini models for chat, embeddings

Groq: high-speed inference of open-weight models (LLaMA 3, Mixtral, Gemma)

Anthropic: Claude series, strong for long-context reasoning

And many more

Hands on

How to use Groq

Metrics in NLP

For classification tasks:

Accuracy: proportion of correct predictions (classification, QA)

Precision / Recall / F1: better for imbalanced classes

However, these metrics **do not work well for text generation tasks**, since they penalize heavily when the generated text uses different words with similar meaning

Reference answer: “The cat sat on the mat.”

Model output A: “The cat is sitting on the mat.”

Model output B: “A feline rests on the rug.”

Metrics in NLP

For generation tasks:

BLEU

ROUGE

Perplexity

Human Score

BLEU

- Measures machine translation quality
- Precision-oriented: Counts how many candidate word pair (ngrams) appear in reference
- Higher BLEU = better translation (closer to reference)

Example:

Candidate: "The cat is on the mat"

Reference: "There is a cat on the mat"

BLEU score ≈ 0.71 (good overlap but missing "There is")

ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

- Focused on recall: how much of the reference is captured by the system
- Commonly used in summarization tasks
- Variants:
 - ROUGE-1: unigram overlap
 - ROUGE-2: bigram overlap
 - ROUGE-L: longest common subsequence

Example:

Reference: "The economy is growing rapidly this year."

Candidate: "The economy grows quickly this year."

ROUGE-1 = high (many word overlaps), ROUGE-L = high (long subsequence match)

Perplexity

- Evaluates language models (LMs)
- Measures how well a model predicts a sequence of words
- Defined as the inverse probability of the test set, normalized by length
- Lower perplexity = model is less “surprised” by the data
- Good for comparing different LMs trained on same task

Thank you!

Questions?